

BERKE ÜNAL

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Education

Oct 2021 – B.Sc. at **Koç University**, Istanbul (Full Scholarship)

- Major in Mathematics (graduated in June 2025)
- Major in Electrical and Electronics Engineering (graduated in June 2026)
- Minor in Physics (graduated in June 2026)

Cumulative GPA: 4.00 / 4.00

Oct 2025 – M.Sc. at **Koç University**, Istanbul (Full Scholarship, Presidential Fellow)

- M.Sc. in Mathematics (expected graduation: June 2027)

Cumulative GPA: 4.00 / 4.00

Publications & Research

- [1] A. Jamneshan, **B. Ünal**, *Quantum uniformity norms are pullbacks of matrix valued uniformity norms*, submitted to QIP, 25 pages, [arXiv:2606.16862](https://arxiv.org/abs/2606.16862)
- [2] U. Tamer, **B. Ünal**, A. Dizdar, Z. Gedik, Ö. Müstecaplıoğlu, *The Quad-C5 Graph: Maximum Contextuality Gap on Eight Vertices*, submitted to Physical Review A, 16 pages, [arXiv:2605.12828](https://arxiv.org/abs/2605.12828)
- Ongoing research with Prof. Christiann van de Ven (Radboud University) on existence of classical KMS states for infinite particle systems based on doi.org/10.1007/s00023-025-01651-2.

Given Talks & Presentations

2026 Summer **ACQ Seminar** organized by Jop Briët (CWI Amsterdam) and Davi Castro-Silva (University of Cambridge)

- Gave a talk titled “The quantum uniformity space within the matrix-valued uniformity space” based on [1].

2025 Fall **Graduate Math Seminars**, Koç University

- Giving a lecture series on the representation theory of C^* -algebras.

2025 Summer **ISem28 Final Workshop**, University of Wuppertal & University of Bonn

- Gave a talk on adjunctions in the category of $\mathbb{Z}[G]$ -modules.

2025 Spring **Collage of Sciences Posters**, Koç University

- Presented a poster titled “Gelfand Representation and Functional Calculus for C^* -Algebras”

2024 Fall **Undergrad Research Day in Mathematics**, Koç University

- Gave a talk titled “Operator Valued Integrals and Holomorphic Functional Calculus”

Scholarship and Awards

- Koç University Presidential Fellowship, the university’s highest-level academic fellowship (2025 – Present)
- TEV Outstanding Success Scholarship (2022 – Present)
- TOG Bezmen Mathematics and Science Scholarship (2022 – Present)
- The Scientific and Technological Research Council of Türkiye (TÜBİTAK), Fundamental Sciences Scholarship (Highest degree) (2021 – 2025)
- Vehbi Koç Honor Award, Koç University (2021, 2022, 2023, 2024, 2025)
- Koç University KUDEB Scholarship (2021 – 2025)

Projects

Further details on these projects are available on my website.

- 2026 Spring The Scientific and Technological Research Council of Türkiye (TÜBİTAK), B92 quantum key distribution protocol using heralded single photons (with Melih Özkurt, Stanford University).
- 2024 Summer Beko Global, Central R&D, designed and prototyped a void fraction sensor to control refrigeration cycles and worked on solutions of the Maxwell equations in the setting of a double helical capacitor.

Teaching

Koç University

- Spring 2026: Grading and office hours for Math303 (Mathematical Methods in Physics, Prof. Ali Mostafazadeh)
- Fall 2025: Grading and office hours for Math320 (Advanced Linear Algebra, Prof. Ali Mostafazadeh)
- Spring 2025: Managing the weekly presentations and serving as session chair Math532 (Graduate Functional Analysis, Prof. Hasan İnci)
- Fall 2023: Gave weekly recitations for Math320 (Advanced Linear Algebra, Prof. Hasan İnci)

Service

- 2024 – 2025 Koç University, Mathematics Department Representative, Student Council
- 2024 – 2025 Koç University, Mathematics Student Mentor, Dean of Students Office
- 2023 – Present Founder of Koç University Blood Donation Platform (300+ members, 40+ donations)

Summer Schools & Workshops

- 2026 Summer The Vision of Sergio Doplicher in Mathematics and Physics – Accademia Nazionale Dei Lincei
- 2026 Summer Hidden Variables in Quantum Mechanics Summer School (Prof. Zafer Gedik) – Village of Sciences
- 2025 Summer Erasmus UNIC Summer School on Mathematics 2025 – Koc University (with ECTS credit)
- 2025 Fall Spring Isem28 Ergodic Structure Theory and Applications, Lecture and Project Phases
- 2024 Summer Erasmus UNIC Summer School on Mathematics 2024 – University of Zagreb (with ECTS credit)
- 2023 Summer Summer School on Positive Operator Semigroups – University of Wuppertal
- 2023 Summer 20th International Workshop on Pseudo-Hermitian Hamiltonians in Quantum Physics
- 2022 Summer Measure Theory School (Prof. Ilgar Eroğlu) – Çakılları Matematik Köyü
- 2022 Summer The “Other” Quantum Mechanics Summer School (Prof. Zafer Gedik) – Village of Sciences

Selected Pre-University Distinctions

- 2019 – 2020 Technology Student Association TEAMS (Tests of Engineering Aptitude, Mathematics, and Science)
- Ranked 1st in the countrywide finals and represented Turkey in the grand finals in the US.
- 2016 – 2021 TEVİTÖL (selective TEV boarding high school for gifted students)
- Graduated as 1st from the program (YKS).
 - I continue to give courses on mathematics and physics to gifted students in Tevitöl.
- 2013 – 2016 ÜYEP (Education Platform for Gifted Students)
- Ranked 1st nationwide in the IQ test designed by Anadolu University.